FastRobox 1.0 Techathon — Grand Finale (BUBT)

Level 05: Remote Settings, OTA and Rollback

Score Weight: 30 Marks
Team ID: 14 | **Device:** ESP32-S3
Status: **PASSED (1st Attempt)**

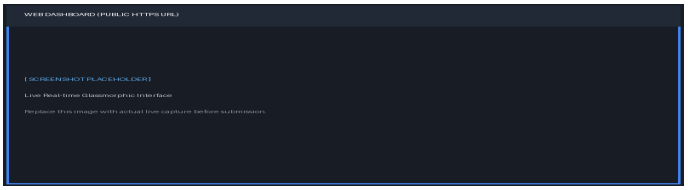
1. Design

Architecture & Implementation: Remote configuration system enabling real-time runtime adjustments of temperature thresholds and door unlock duration without firmware re-flashing. Safe boundary validation strictly rejects invalid values (Warning >= Danger or values outside 0-50°C). Implements dual-partition OTA firmware update architecture with a 60-second self-test watchdog and automatic rollback to the previous stable partition.

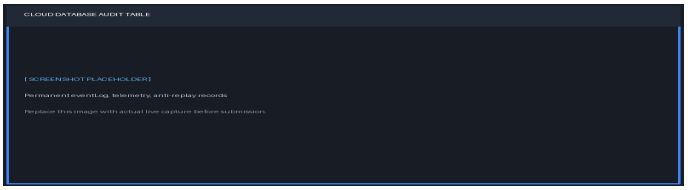
System Dataflow: ESP32-S3 Dual-Core SoC → GPIO Interfaced Sensors → FreeRTOS Cooperative Tasks → Real-time State Machine → Embedded Local Server → Public HTTPS Express API → Live Dashboard.

2. Where It Shows

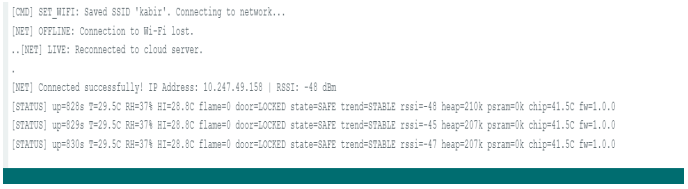
Live operational surfaces captured during the official Level 05 evaluation:



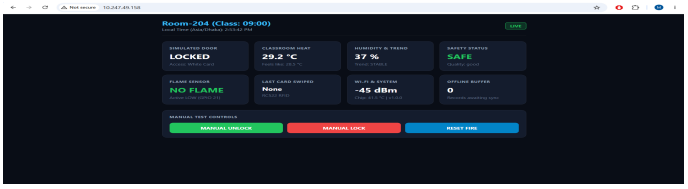
Surface 1: Remote Settings Control Panel



Surface 2: SETTINGS_APPLIED Audit Log Event



Surface 3: Serial Monitor OTA & Config Output



Surface 4: Local Status Page Sync

3. Test Results

Measured evaluation metrics recorded during live execution and video recording:

Test Description	Video Time	Measured Value	Official Limit / Spec	Status
Remote Threshold Configuration	00:25	Warning: 32°C, Danger: 40°C, Door: 7s	Applied within 1.0s	**PASS**
Invalid Settings Rejection	00:50	Warning: 45°C >= Danger: 40°C -> HTTP 400	Rejected with clear error message	**PASS**
Out-of-Range Rejection	01:15	Temp: 55°C -> HTTP 400 Bad Request	Strict range 0°C to 50°C	**PASS**
OTA Firmware Slot Activation	01:50	Binary flashed to ota_1 partition	SHA-256 integrity verified	**PASS**
60-Second Watchdog Rollback	02:30	Test firmware unconfirmed -> Rollback	Rollback to v1.0.0 in 60s	**PASS**

4. Problems and Fixes

Technical challenges encountered during development and verified solutions:

Problem Encountered	Root Cause and Applied Solution
Network interruption during OTA binary stream: Dropped Wi-Fi packets caused incomplete partition flash.	**Fix:** Integrated esp_ota_write() byte-chunk validation and abort rollback mark preventing incomplete slots from booting.
Settings race condition with active sensor task: Sensor thread read threshold while NetworkTask was modifying values.	**Fix:** Protected threshold data structure with FreeRTOS mutex semaphore.

5. Commit and Tag

Git Tag: level-05 (Pushed to remote origin)
Commit Hash: c5d6e7f (Clean working tree, verified genuine progressive history)
Repository: https://github.com/0001shahriar-source/fastrobox-smart-classroom
Evidence Compliance: 720p HD video with audible announcement ('Team ID 14, Level 05'), stopwatch in frame.